PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

umhackathon@um.edu.my

© UMHackathon 2026

Email: umhackathon@um.edu.my

Table of Contents

1. Project Overview 1
2. Background & Business Objective 1
3. Product Purpose 2
4. System Functionalities 2
5. User Stories & Use Cases 3
6. Features Included (Scope Definition) 3
7. Features Not Included (Scope Control) 3
8. Assumptions & Constraints 4
9. Risks & Questions Throughout Development 4

© UMHackathon 2026

Email: umhackathon@um.edu.my

Project Name: StockSense

Version: 1.0

1. Project Overview

- **Problem Statement:** Supply chain management often suffers from inaccurate stock forecasting, delayed supplier communication, and lack of proactive decision making. Businesses frequently experience overstocking or stockouts due to poor demand prediction and manual monitoring. These inefficiencies increase operational costs and reduce business agility.
- **Target Domain:** AI powered Supply Chain Automation & Inventory Management
- **Proposed Solution Summary:** StockSense is an AI powered supply chain automation system that predicts stock demand, automates supplier communication, and generates weekly stockholder newsletters. Using GLM5 from Z.AI, StockSense analyses historical demand forecasting data and simulates intelligent supply chain decisions through n8n workflows.

2. Background & Business Objective

- **Background of the Problem:** Traditional supply chain management relies heavily on manual forecasting and reactive decision making. Businesses often depend on spreadsheets or static reporting tools to monitor stock levels. These methods are inefficient and prone to human error. When demand unexpectedly increases, companies experience stock shortages. Conversely, when demand decreases, businesses face excess inventory and increased storage costs.

© UMHackathon 2026
Email: umhackathon@um.edu.my

Moreover, supplier communication is often inefficient and inconsistent. Businesses typically maintain a fixed supplier list and rarely explore alternative suppliers unless a disruption occurs. This lack of flexibility reduces resilience within the supply chain.

Another key issue is poor stakeholder communication. Stockholders and business managers often lack clear insights into inventory performance, future demand trends, and procurement decisions.

- **Importance of Solving This Issue:** Improving supply chain intelligence can significantly enhance business performance. By predicting demand and automating supplier communication, companies can reduce operational costs and improve responsiveness. Automated reporting also ensures transparency and better strategic planning.

  StockSense addresses these problems by combining artificial intelligence with workflow automation to create a proactive and intelligent supply chain system.

- **Strategic Fit / Impact:** StockSense demonstrates the practical application of AI in business automation. The system aligns with AI driven operational intelligence and showcases how large language models such as GLM5 can enhance enterprise workflows. This project also highlights the integration of AI models with automation platforms such as n8n, making it scalable and adaptable for real world business environments.

3. Product Purpose

3.1. Main Goal of the System: To automate supply chain decision-making using AI based demand forecasting and intelligent workflow automation.

3.2. Intended Users (Target Audience):

- Supply Chain Managers
- Business Owners
- Warehouse Managers
- Inventory Managers
- Procurement Teams

4. System Functionalities

4.1. Description: StockSense operates as an AI powered supply chain automation platform. The system processes demand forecasting data to identify patterns and predict future inventory requirements. When the system detects potential stock shortages, it automatically generates supplier recommendations and prepares communication workflows.

In addition, StockSense generates weekly newsletters summarising product demand, inventory status, and forecast insights. These newsletters provide stakeholders with a clear overview of business performance.

The system is deployed using GLM5 from Z.AI for intelligence and n8n for workflow automation. The demand forecasting dataset is used for simulation and model testing.

4.2. Key Functionalities:

- **Stock Prediction**: Predict future demand and suggest restocking timing
- **Supplier Contact Automation:** Identify existing suppliers, if none, suggest alternative ones to generate automated communication between supplier and operator
- **Weekly Newsletter**: Generate weekly reports to stockholder

© UMHackathon 2026

4.3. AI Model & Prompt Design

4.3.1. Model Selection: StockSense uses GLM5 from Z.AI. The model is used for demand analysis, prediction, supplier recommendation, and newsletter generation. GLM5 is chosen for its strong reasoning ability and support for multi step decision making.

4.3.2. **Prompting Strategy:** StockSense uses multi step prompting with multiple AI agents. Each agent handles a specific task, while a main orchestration agent controls the overall workflow. The orchestration agent manages the process flow. It decides when to trigger each agent and passes outputs between them.

The demand prediction agent analyses data and predicts future stock levels. The conversation automation agent generates supplier messages and suggests alternative suppliers. The newsletter agent generates weekly reports based on stock data and prediction.

GLM5 supports multi step reasoning, which allows the system to break tasks into smaller steps and produce more consistent outputs.

4.3.3. **Context & Input Handling**: StockSense processes structured demand forecasting datasets. When handling large datasets, the system performs batch processing and summarization. If the input size exceeds the model's context limit, the dataset is divided into smaller chunks and processed sequentially.

This ensures the system can handle large volumes of data efficiently without losing critical information.

4.3.4. **Fallback & Failure Behavior:** If the AI model produces off-topic or inaccurate outputs, the system automatically retries using structured prompts. If the issue persists, the system generates fallback responses and marks the output as simulated. This ensures reliability and prevents incorrect automation actions.

© UMHackathon 2026

5. User Stories & Use Cases

- As a warehouse manager, I want to predict stock demand so that I can prevent shortages.
- As a business owner, I want automated supplier contact so that I save time.
- As a stockholder, I want weekly reports so that I understand business performance.

- **Use Cases (Main Interactions):**

  Prompt to UI Flow: The system receives the dataset, the orchestration agent sends it to the demand prediction agent to analyse data and generate stock predictions, if low stock is detected the conversation agent suggests suppliers and generates a contact message. On a scheduled basis, the newsletter agent collects data and generates a weekly summary newsletter.

  Flow Generation: The orchestration agent controls all flows. It decides which agent to trigger based on system conditions. It ensures outputs from one agent are passed correctly to the next step. This allows the system to move from prediction to action to reporting in a structured way.

- **Use Case Diagram:** *(Optional but encouraged for participants to visualize the interaction)*.

6. Features Included (Scope Definition)

- AI stock demand prediction
- Supplier recommendation system
- Automated newsletter generation
- n8n workflow automation
- Simulation environment

© UMHackathon 2026
Email: umhackathon@um.edu.my

- Dataset-based forecasting

7. Features Not Included (Scope Control)

- Real time supplier API integration
- Production ready deployment
- Real inventory system integration
- Multi-company support

8. Assumptions & Constraints

- **LLM Cost Constraint:** During the development, the system consumed a total of 880k tokens over 239 requests
- **Technical Constraints:** The current version is limited to processing structured data within google sheets
- **Performance Constraints:** To ensure model accuracy and avoid context window limits, the system enforces a 1500 product batch limit per regional forecasting run
- **User Input:** The autonomous workflow includes a feature that requires human approval in gmail for orders exceeding a set financial threshold

9. Risks & Questions Throughout Development

- Coordination between agents: errors can occur if outputs are passed incorrectly between agents.
- Inconsistent output: Different agents may generate results in different formats. This can cause issues when the orchestration agent tries to combine them
- Multi step reasoning: If one produces an incorrect result, the error can carry over to the next steps.

© UMHackathon 2026